NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R21.4 Guangzhou Urban Wind GPU-LBM Reproduction Feasibility (formerly L22.1)

Group Digital Wind Tunnel · R21.4 Urban Pedestrian-Level Wind LBM (LUW)

Status Promoted

Period 2026-10-13 00:15 — 2026-10-15 17:40

Stage conclusion

Project report

Guangzhou Urban Wind Field GPU-LBM (LatticeUrbanWind) Reproduction Feasibility

> **Promoted (10-15 16:1x)**: Approved by Liang Hu at 10-15 14:00; formal project "Urban Wind Field GPU-LBM (LUW)" opened: project directory, Root Lead, and bot `luw`. This page, 17 decision records, handoff documents, and project files (2,543 files, 168 MB) have been copied over; the new project is authoritative hereafter and this page is no longer updated.

Objective

Determine whether Han et al. 2026 (DiBE 101057; arXiv 2607.04516v2; code LUW-v2.8) is reproducible and whether it can be put to use for our work. Deliverables: the assessment table below, a minimal-case execution log, and a one-line conclusion (worth continuing / Shelved / recommend handing over to the Digital Twin Wind Tunnel line).

Current Status

10-15 17:40 Promoted: The formal project (bot `luw`) has been opened and has received the handoff (materials and project page transferred); this page is now closed, and all subsequent work proceeds under the new project. **10-15 14:00 Liang Hu approved promotion**: promotion proposal and handoff checklist (high priority): new project directory, Lead, and bot `luw`, notifying me for handoff once set up and smoke-tested; once live, mark this page Promoted and link to it. **10-15 09:30 Equilibrium log-law verification wrap-up**: the wall law cannot sustain an equilibrium log-law on a 1 m grid (20.6% low at 2 m); code is correct, physics falls short, prototype frozen and paused. **10-15 00:55 Ground wall-model prototype paused** (decision): implementation is correct, reducing empty-fetch near-ground decay from ~40% to ~22% but still falling short of the target, while Case E error increased rather than decreased—the ground boundary layer is not the primary error source in Case E; materials handed over to the Digital Twin Wind Tunnel line for read-only reference (their 10-15 receipt: agreed with conclusion and pause, to serve as their wall-sensitivity starting point once unfrozen). **10-14 20:30 Round 13 design reviewed and cleared** (decision, delegation queued, ≤15 CU; do not proceed to the urban case if the empty-fetch gate fails). **10-14 20:15 Round 12 returned**: near-wall underestimation is not an interpolation artifact (removing solid corners reduced MAE by only 0.012), and probe points are highly sensitive to 1–2 m wall offset; do not alter probe coordinates or attribute the discrepancy to wall mechanics on this basis; remaining work focused on Round 13 ground wall model (`ideas` designing, Lead to review `DESIGN.md` first). **10-14 Liang Hu approved proceeding**: Round 12 near-wall sampling sensitivity and Round 13 ground wall-model prototype (design first) dispatched; Digital Twin Wind Tunnel Lead replied: no duplication, proceed freely; requested `DESIGN.md` and Case E/A validation results, noting that their "first-layer log-law stress + retained bounce-back" approach had almost no effect—the wall model must replace bounce-back and include empty-fetch profile/turbulence-intensity maintenance and ground momentum budget checks (forwarded to `ideas`). Previously: **stage summary submitted to Liang Hu**, awaiting Liang Hu's decision on next steps; LBM software comparison table finalized; stage summary written thereafter. Ground boundary layer (wall model) was the next candidate, requiring division of labor with the Digital Twin Wind Tunnel line as decided by Liang Hu. Previously: **10-14 15:00 All configuration-level tests completed** (decision). Round 9 Case A inflow turbulence configuration proved clearly effective. Earlier in Round 9: Liang Hu asked "why is it out of scope," and Lead corrected: inflow turbulence can be tuned purely via configuration (`vk_inlet_*` options) without modifying source code; only ground roughness / wall models require source code changes. Dispatched Case A inflow turbulence configuration experiment (≤3 CU).

Previously: **10-14 01:30 Round 7 completed, returned to Shelved** (decision): on the exact same mean field, switching to the wind-tunnel scalar wind speed definition $S = (S_{\text{CFD}}^2 + 2k)^{1/2}$ (`C05_5` appendix,

where k includes only resolved scales) improved post-redevelopment 80-point MAE from 0.144 to **0.111**, mean ratio from 0.81 to **0.95**, with $r = 0.82$; the remaining gap is concentrated at 13 low-speed points (where the simulation still reaches only 39% of wind-tunnel values, with 4 points touching solid cells during interpolation), and near-ground inflow decay persists (2 m wind speed drops from 103% to 41% along the upwind fetch). Further investigation of these two issues requires modifying source code (ground roughness / wall model) and belongs to formal research.

Previously: **10-14 01:30 Resumed Shelved status** (decision). After Liang Hu noted that "a substantial gap remains," Round 7 was conducted: because the wind tunnel measures scalar wind speed, comparing via the official AIJ approximation $S_{\text{exp}} = \sqrt{S_{\text{CFD}}^2 + 2k}$ on the same mean field improved 80-point MAE from 0.144 to **0.111** and simulation/wind-tunnel mean ratio from 0.81 to **0.94** ($r = 0.82$). Most of the overall underestimation stemmed from the comparison metric definition. What remains: 13 wake low-speed points reach only 39% of wind-tunnel values; 2 m wind speed drops to 0.41 within 42 m of the inlet; extending the upwind flat fetch to 509 m lowers the mean ratio by a further 0.11 (under identical timing settings). These belong to near-ground solver treatment and will not be dug into further in the trial ground. **Still not termed a "reproduction"** (not submitted for review on these new numbers; the wording rules from the review verdict remain in force).

Below is the earlier 10-14 shelving record (superseded numerically by the paragraph above): **10-14 Shelved (question answered, paused)**, per the review verdict. - **Conclusion (post-review wording)**: The public code `LatticeUrbanWind` HEAD `dd438c1` (which is not identical to the `LUW-v2.8` cited in the paper and lacks the immersed boundary described in Supplementary Material S5) runs on a Colab A100; under the paper's west wind and 1 m resolution, on a self-defined ~500 m horizontal domain (49 m upwind flat fetch) with 15.9 m normalization and $(+1,+1,0)\text{ m}$ registration offset, the 80-point post-redevelopment west-wind MAE is 0.1425—**numerically comparable** to the paper's 0.143—but the simulated mean remains 19% below the wind tunnel (with a 509 m upwind flat fetch, MAE is 0.157 and the mean is 30% low). **Not termed a "reproduction."** - Underestimation in the first three rounds (~3× on the coarse grid, ~2× at 1.86 m) was obtained under conditions unequal to the paper (wind direction, grid resolution, domain size, and averaging window differed simultaneously), with individual factor contributions unseparated. - Observation: under these settings, 2 m wind speed over the upwind flat ground retains only 33–53% of the inflow value while 15.9 m is largely preserved; hypothesized to be related to the smooth flat ground lacking roughness, without a controlled test. - Public code output coordinates are shifted by $(+1,+1,0)$ grid cells relative to the STL (confirmed empirically via solid mask; code mechanism not fully closed). - Total shared Colab usage ~20 CU; two CRC launch attempts failed (0 SU).

Why Shelved

The question the Lab set out to answer—"can it run, and can we use it?"—has been answered: it runs, and under the paper's settings its metrics reach the paper's level; however, it is not the paper's version (lacking the immersed boundary), Clause 5 of the license restricts publishing results from modified versions, and the Digital Twin Wind Tunnel line does not use it for production. Digging deeper (scalar wind speed, rough ground, code version) belongs to formal research rather than the trial ground.

Restart Conditions

- The Digital Twin Wind Tunnel line needs AIJ Case E as an external benchmark (official data, registration, offset correction, and metric scripts are all ready; a Round 5 configuration run takes ~12 minutes and ~1 CU);
- The authors release `LUW-v2.8` or a version containing the immersed boundary;
- Observation-driven boundaries need to be tested and multi-station, multi-height observations become available.

Below is the historical record: **10-13 Reopened (shelving retracted)**: Liang Hu asked "why can't we reach the paper's performance," and upon re-checking the paper the Lead found that AIJ Case E in the paper used **west wind and a 1 m grid**, achieving MAE 0.143 (Han 2026 arXiv v2 pp. 11–12, Fig. 1), whereas our first three rounds used north wind and 3.39 / 1.86 m grids—meaning the earlier "unreproducible" conclusion was drawn under unequal conditions. Dispatched Round 4 to re-run under the paper's settings (≤15 CU hard cap; crop domain to paper dimensions if an 80 GB card is unavailable), per decision.

Below is the pre-retraction assessment, retained for reference: **10-13 Shelved (retracted)** (decision). One-line conclusion: **The LUW solver runs easily on our Colab A100, but under its bundled default settings it cannot reproduce pedestrian-level wind speeds in AIJ Case E; the borrowable methodology (leave-station-out validation + pure interpolation baseline) has already been absorbed by the Digital Twin Wind Tunnel line, so further investment in the Lab is not warranted.** Three rounds spent ~9.4 CU in total (2.21 + 0 + 7.19).

Why Shelved

1. Running it works out of the box: 3 minutes on a 3.39 m grid and ~25 minutes on a 1.86 m grid via configuration changes alone.
2. Mismatch with the wind tunnel is largely diagnosed: across the 80 probes, mean 2 m wind-speed ratio is 0.113 (3.39 m grid) → 0.217 (1.86 m grid) vs. wind-tunnel values of 0.34 (pre-redevelopment) / 0.42 (post-redevelopment); on the 1.86 m grid, MAE is 0.17/0.22 and $r = 0.41/0.46$. **The primary cause is near-ground inflow decay before reaching the buildings**: at $y = 1600\text{ m}$ upstream of the building cluster, 2 m wind speed is already down to 42% of the inflow profile (1.24 vs. 2.96 m/s), whereas 15.9 m remains normal (4.05 vs. 3.93). This is a ground-boundary and boundary-layer issue, not a coordinate registration error (ruled out in Round 2). Grid refinement also lengthened the averaging window from 87 s to 571 s, so their contributions cannot be separated.
3. Digging deeper requires modifying LUW's ground boundary or inflow (source code changes), which triggers Clause 5 of the license (publishing results from modified code requires open-sourcing the modifications); moreover, the Digital Twin Wind Tunnel line does not use the LUW solver and is building its own precursor inflow (CRC 1535416), leaving no downstream recipient for continued investment here.
4. What is borrowable from the methodological chain (leave-station-out validation must include a pure interpolation baseline, Han card p. 39 Table 6) has already been absorbed by that line; observation-driven boundaries lack multi-station, multi-height observational data (Guangzhou lidar data unavailable, p. 44).

Restart Conditions

- The Digital Twin Wind Tunnel line needs an external GPU-LBM urban-canopy benchmark (probe registration, comparison scripts, and wind-tunnel data parsers for AIJ Case E are all prepared and ready for direct reuse); or
- The LUW authors release a new version with a wall model / rough-ground inflow; or
- Usable multi-station, multi-height urban observations (such as CROCUS Chicago) become available to test observation-driven boundaries.

Information for the Digital Twin Wind Tunnel Line (Forwarded via Root Lead)

In FluidX3D-family codes, the default no-slip flat ground combined with a logarithmic inflow profile decays 2 m wind speed to ~40% of its inflow value over an upstream fetch of ~1.6 km on a ~2–3 m grid, while 15.9 m wind speed is largely maintained. This is analogous to the "streamwise decay of wind speed and turbulence intensity downstream of the inlet" observed in their T3 short test and serves as a reference point. Based on empirical data from this line rather than literature.

Pre-Launch Findings by Root Lead (Not Duplicated)

- The Digital Twin Wind Tunnel Urban Field sub-project already evaluated the LUW **preprocessing chain** (SHP/DEM/STL/GPU voxelization; local synthetic-domain short test passed, full domain not run) and noted Clause 8 of the LUW license (funding source restrictions; funding classification pending Liang Hu's decision).
- Han 2026 literature card: leave-station-out validation gives LBM–LES MAE of 0.38–0.53 m/s (p. 38 Table 5); on the exact same pairs, pure Kriging interpolation achieves MAE 0.36–0.37, outperforming LBM–LES at 0.46 (p. 39 Table 6); validates only mean horizontal wind over a single period (pp. 39–43).
- Consequently, the expected gap for this project is: whether the LUW **solver proper** can run on a Colab A100 (using its bundled minimal case or AIJ), and the substitutability of observation-driven boundaries and conservation corrections within its methodological chain.

Reproducibility Assessment Table

Component	Feasibility	Obstacles	Required Resources
License (LUW custom license + FluidX3D)	**Runnable for internal evaluation**; funded research requires prior funding-classification decision	Clause 8: for funded research, the funder must be a government or university funding agency (e.g., NSF); funding from other government departments or non-commercial bodies requires prior email permission from the author; commercial funding is strictly prohibited. Clause 5: publishing results produced by modified versions requires open-sourcing the modified code. Clause 3 prohibits military use	Liang Hu to determine funding classification (shared pending item with the Digital Twin Wind Tunnel line)
Solver compilation & execution (Colab A100, OpenCL)	**Feasible (empirically verified)**	Official install script fails when creating a `venv` on Colab, bypassed via `uv`; `make Linux` compiles cleanly	Setup plus one solve takes ~0.5 h, 2.2 CU
AIJ Case E (Niigata) minimal case	Runs; **cannot reproduce under default settings**	Simulated 2 m wind-speed ratio is 0.22 vs. wind-tunnel 0.34/0.42 (1.86 m grid); primary cause is streamwise decay of near-ground inflow to 42% of input; registration error ruled out; unclear from repo whether STL corresponds to pre- or post-redevelopment	9.4 CU total; finest grid on a single 40 GB card is ~1.64 m
Guangzhou lidar observations	No	Paper states data is available from the corresponding author upon "reasonable request" (Han card p. 44); external outreach rests with Liang Hu	—
Building geometry substitution	Feasible	Digital Twin Wind Tunnel line has already built OKC 5 m geometry (v2) from 3DEP lidar nDSM; LUW skips buildings missing height attributes, so missing-height auditing must be added separately	Not performed in this project

Component	Feasibility	Obstacles	Required Resources
Observation-driven boundary (level-by-level 2D ordinary Kriging → Delaunay/kNN mapping onto lateral faces, held constant within each analysis step, with synthetic von Kármán turbulence superimposed at the inlet)	Algorithmically feasible; data is the bottleneck	Requires multi-station, multi-height profiles: acquisition channel for JU2003 unknown; CROCUS Chicago 2024 is open (listed as a candidate in the Digital Twin Wind Tunnel review)	Cannot test without observational data
Conservation correction (equilibrium-velocity open boundaries on four sides and top + damping buffer + global mass-flux correction)	Borrowable concept	Our in-house v7 uses body-force inflow + sponge layer with a different architecture; license forbids copying source code (Digital Twin Wind Tunnel line has decided to use it for design reference only)	—
Leave-station-out validation	Directly borrowable practice	Key lesson: pure Kriging interpolation achieves MAE 0.36–0.37 m/s, lower than LBM–LES at 0.46 (Han card p. 39 Table 6), so a pure interpolation baseline is mandatory; input and held-out stations belong to the same network and are not fully independent (pp. 39–43). Recorded under "Applicable to" on the Digital Twin Wind Tunnel line's Han card	Already absorbed by that line

Methodological chain basis: Han 2026 literature card (arXiv v2 pp. 7–10, 15–19, 38–44).

Parameter Source Table (AIJ Case E; Rule Set by Liang Hu on 10-13)

Parameter	Value	Source or "Assumption"	Places Checked
Wind direction	West wind (paper); LUW `angle=270`	Han 2026 arXiv v2 p. 12; source `setup.cpp` L2685–2689	Paper, source code
Grid resolution	1 m uniform	Han 2026 p. 12	Paper

Parameter	Value	Source or "Assumption"	Places Checked
Computational domain	Official AIJ intercomparison: ~500×500×300 m; wind-tunnel turntable diameter 400 m (full scale), 1/250 scale; geometry in Han Fig. 1(b) is also this 400 m disk; LUW example uses 2022×1996×270 m	`C05_5` Table 1; Gemini report citing official Japanese documentation Tables 1–2 (pending Lead spot check of original text); Han p. 11 Fig. 1(b)	Paper main text, S1–S8, repo, official AIJ materials
Reference wind speed U_{ref}	Inflow wind speed at 15.9 m (Niigata meteorological station anemometer height), ~3.93 m/s	Official `.xls` `Results` sheet row 2 note; all 2,560 wind-speed ratios in `.xls` are integer multiples of `0.019820405` (noted in Gemini report, verified by Lead), meaning raw values were recorded in units of $0.01\,U_R$ and converted to 15.9 m. **Han 2026 uses this exact definition**: in Fig. 1(d) (p. 11), wind-tunnel probe 37 is ~1.18 and probe 16 is ~0.55, matching 1.169 and 0.535 in the `.xls` post-redevelopment west-wind column (verified by Lead from figure). Note: figures in the original AIJ paper normalize by "inflow wind speed at the same height (2 m)" (Yoshie 2007 JWEIA 95 p. 1569; `C05_5` main text), yielding values 1.33× those in `.xls`; the two must not be mixed	Paper, S1–S8, official `.xls`, `C05_5`, Yoshie 2007, Gemini report
Inflow profile	Official `Inflow` table ($Z_R = 250\text{ m}$, $U_R = 7.8\text{ m/s}$, power-law exponent ~0.25)	Official `.xls` `Inflow` sheet; `C05_5` main text; consistent with repo `profile.dat`	Same as above
Ground boundary	Official: logarithmic law, $z_0 = 0.024\text{ m}$; LUW: voxel bounce-back, zero roughness	`C05_5` Table 1; source code	Same as above
Wall treatment	Paper: regularized delta-kernel immersed boundary (S5); public code: no such implementation	Supplementary PDF pp. 6–10; search across all three solver directories	Paper, S1–S8, source code

Parameter	Value	Source or "Assumption"	Places Checked
Building configuration	Post-redevelopment (includes 60 m proposed building A and 18 m buildings B and C); Han 2026 also compares against the post-redevelopment west-wind column (see Fig. 1(d) check above)	`C05_5` main text Niigata section; buildings A/B/C on separate layers in official DXF (Gemini report); max STL height 60 m	Same as above
Wind-tunnel measurand	Scalar wind speed (hot-wire/thermistor anemometer), model 8 mm = full-scale 2 m	`C05_5` same section	Same as above; LUW outputs the magnitude of the mean velocity vector—**a different definition that biases wake regions low**
Code version	Paper: `LUW-v2.8`; public repo has no such tag, using HEAD `dd438c1`	Paper p. 44; GitHub refs	Repo, releases, tags

Trials and Results

Date	Action Taken	One-Line Result
10-15 09:28	Equilibrium log-law empty-fetch test (~2 CU, final)	After switching the inlet to a $z_0 = 0.024$ logarithmic profile, the first-layer wall law still failed to maintain the profile: 20.6% low at 2 m (vs. 37% low without the model) and 3–7% fast at 5–20 m, a classic log-layer mismatch; model u_* is ~0.79 of the specified value. Verdict: code is correct, physics falls short, prototype remains paused
10-15 00:46	Round 13 Continuation 5: height datum, near-wall damping, Case E diagnosis (~4–5 CU)	Wall sits at physical 0.5 m (2 m probe is 1.5 m above the wall, ~6% low), no registration offset; near-wall damping had almost no effect (2 m ratio 0.77–0.79); Case E with first-layer wall law: scalar MAE rose from 0.116 to 0.149 and mean ratio from 1.06 to 1.20, improving near walls but overshooting in open areas. Lead decided to pause the wall model and hand materials to the Digital Twin Wind Tunnel line for reference

Date	Action Taken	One-Line Result
10-14 23:50	Round 13 Continuation 4: `R10a` inflow, 200 m wide empty fetch (~2 CU)	2 m wind speed / inlet at $x = 200\text{–}400\text{ m}$: 0.56–0.65 without model, 0.75–0.79 with first-layer wall law, 0.69–0.79 with second-layer reference; stabilizes from 100 m onward without further decay, but stays below the 0.85 gate; effective roughness ~0.2 m (vs. 0.024 specified). Dispatched: check height datum, near-wall subgrid damping variant, and one diagnostic Case E run
10-14 23:12	Round 13 Continuation 3: pairwise swap + first empty-fetch run (~3–5 CU)	Gate 0 passed (zero-force conservation, point-by-point match with HEAD, wall force closure within 4%). Empty fetch at $x = 200\text{–}400\text{ m}$ gave 2 m wind speed / inlet of 0.43–0.48 without model and 0.55–0.60 with model, while 15.9 m remained stable (<10%); failed gate. Lead found the empty-fetch run used weak inflow turbulence (TI 0.05) and a narrow 64 m domain, and the criterion did not account for the inflow profile not being a $z_0 = 0.024$ log profile (equilibrium expectation ~0.96); dispatched re-run
10-14 22:14	Round 13 Gate 0: two specular-reflection attempts in LUW kernel (~3 CU)	Under zero force, x-momentum dropped by 52%/89%, failing Gate 0; Lead inspected the diff and identified an incorrect population mapping (using distributions incoming from above). Correct procedure: pairwise swap $9 \leftrightarrow 16$ and $11 \leftrightarrow 18$ after `load_f`. Dispatched third attempt followed by empty fetch
10-14 21:46	Round 13 Gate 0 mini-kernel (Colab, ~1 CU)	Zero-force specular reflection conserved mass and momentum to 10^{-6} (passed); with wall force applied, momentum increment reached only ~30% of applied force—Lead traced this to a test-harness bug where the mini-kernel Guo equilibrium velocity omitted $F/2$, not a flaw in the wall-model scheme; Case A z_0 fitted from official 2–8 m inflow as $2.95 \times 10^{-4}\text{ m}$ (same order of magnitude as secondary-source 1.8×10^{-4}). Dispatched: fix harness, port to LUW kernel for Gate 0 and empty fetch

Date	Action Taken	One-Line Result
10-14 21:20	Round 13 implementation Round 1 (0 CU)	Performed source verification and static pre-checks only, without GPU execution; found secondary-source value $z_0 = 1.8 \times 10^{-4}\text{ m}$ for Case A (Sim 2015 citing Mochida 2002); Lead instructed continuation: run Gate 0 on Colab with local specular reflection and cross-check Case A z_0 by fitting the official inflow profile
10-14 20:11	Round 13 ground wall-model design (0 CU)	Specular reflection + first-layer log-law body force replacing ground bounce-back, estimated at 135–270 lines of source code; cleared conditionally by Lead (verify y_1 and Case A z_0 first, pass empty-fetch gate)
10-14 20:00	Round 12 near-wall sampling sensitivity (`R10a` re-run, changing sampling only, ~1.7 CU)	Excluding solid corners from interpolation reduced MAE on the four near-wall probes only from 0.345 to 0.333 (mainly at probe 52); probe 50 lies entirely in fluid cells and is nearly stagnant inside a narrow alley; shifting probes 51/52/54 1–2 m away from the wall reduced their MAE from 0.32 to 0.24 to 0.15, but shifting control probes increased their error from 0.10 to 0.12/0.14. Conclusion: underestimation is not an interpolation artifact; probes are highly sensitive to wall distance and cannot be "fixed" by altering sampling, nor can wall mechanics be inferred from this
10-14	Round 11 local geometry diagnosis (0 CU)	Underestimated probes 50/51/52/54 all lie within 1 m of a wall where sampling touches solid cells; overestimated probes 23/25/37/17 lie 5–10 m from walls in pure fluid cells with no geometric distinction from control points; the two clusters have different causes
10-14 14:53	Round 10 Case E four configuration tests (L4, ~6–11 CU)	Vector MAE 0.140–0.149, scalar MAE 0.115–0.119 (vs. Round 7 0.144/0.111); inflow turbulence raised near-ground inflow from 41% to 52% without changing overall metrics; mass-flux correction is a no-op; domain height and averaging window affected MAE by ±0.005; point-wise biases remain concentrated at fixed probes (50–54 low, 23/25/37/17 high) and are insensitive to configuration

Date	Action Taken	One-Line Result
10-14 11:00	Round 9 Case A modifying inflow turbulence only (roof-equivalent TI 0.147, windward-face injection only; 0.8 CU)	TKE FAC2 improved to 0.69/0.93/0.97 (paper: 0.31/0.90/0.90), $r = 0.37/0.45/0.36$ (paper: 0.64/0.88/0.53); mean wind speed also improved: FAC2 0.98/0.93/0.99, $r = 0.91/0.98/0.99$. Amplitude and injection face were changed together without separation
10-14 10:40	Round 8 AIJ Case A official measurement planes (4/40 m + vertical section, 2 m grid, ~2.5 CU total)	Mean wind speed: 4 m plane FAC2 0.95 (paper: 0.74), 40 m 0.83 (0.93), vertical section 0.85 (0.81), $r = 0.81\text{–}0.95$, matching or outperforming the paper; TKE: FAC2 0.35/0.66/0.77 (paper: 0.31/0.90/0.90), $r = 0.24/{-0.01}/{-0.21}$ (paper: 0.64/0.88/0.53), markedly worse. Inflow turbulence used default 5% uniform synthetic turbulence and k included only resolved scales; q and NRMSE definitions labeled as assumptions
10-14 10:10	Round 8 AIJ Case A initial run (0.8 CU)	Solve succeeded, but planes were sampled at $z/H = 8/80\text{ m}$ as written in the paper whereas official measurement planes are at $z/b = 4/40\text{ m}$, invalidating the result; A100 reallocation failed (platform). Lead judged z/H in the paper to be a typo and dispatched a re-run at 4/40 m (using L4/T4 when A100 is unavailable)
10-14	Point-by-point digitized comparison against paper Fig. 1(d) (Lead)	Digitized points vs. wind tunnel give MAE 0.144 (paper reports 0.143, confirming digitization reliability); paper LUW mean ratio is 1.06 (high) with $r = 0.57$; our vector metric gives 0.81 (low) with $r = 0.82$; the two simulations differ from each other by MAE 0.16–0.20; probes 47, 50, and 74 are near 0 in both
10-14 01:23	Round 7: 500 m domain, 31,000/20,000 steps, TKE output enabled	Same field $S_{\text{CFD}} \to S_{\text{exp}}$: MAE 0.144 → 0.111, mean ratio 0.81 → 0.94, $r = 0.82$; under identical timing settings, the long-fetch domain (`r6b`) has a 0.11 lower mean ratio; ~1.7–3.6 CU (two concurrent executors, attribution pending platform check)
10-13 21:37	CRC 1536823 (west wind, 1 m, cropped domain 1200×700 m)	Launch failed with exit code 127: batch environment did not initialize `module`; no fields or metrics produced; re-running in a new root

Date	Action Taken	One-Line Result
10-14 01:23	Round 7: Round 5 configuration + TKE output, 256 s averaging	Post-redevelopment MAE: vector magnitude 0.144 → scalar definition 0.111; mean ratio 0.81 → 0.95; 13 low-speed points still at 39% of wind tunnel; ~2–4 CU
10-13 23:38	Round 6: upwind flat fetch extended 49 → 509 m (otherwise identical to Round 5, 256 s averaging window)	Post-redevelopment MAE 0.157 (offset-corrected) / 0.163, $r = 0.82$, mean ratio 0.70; over the flat fetch (109–459 m ahead of buildings), 2 m wind speed is 33–53% of inflow while 15.9 m is 99–104%; 5.97 CU
10-13 19:41	Round 5: official-scale 500 m domain, west wind, 1 m	Post-redevelopment MAE 0.1425 (with mask-verified $+1$ grid offset in x,y) / 0.148 (unshifted), $r = 0.80$, mean ratio 0.81; comparable to paper's 0.143; 1.11 CU
10-13 11:47	Round 4 west wind, 1 m (three Colab attempts)	Verified west wind is `angle=270`, inflow data matches AIJ table, and 1 m grid launches; all three runs were reclaimed at 29–36% with no output; ≤15 CU total
10-13	Round 3 refined to 1.86 m (7.19 CU)	Simulated mean 0.113 → 0.217, MAE 0.24 → 0.17 (pre-redevelopment), $r = 0.25 \to 0.41$; upstream 2 m wind speed decayed to 42% of inflow while 15.9 m remained normal; `--high-mem` not allocated, 1.64 m grid aborted over budget
10-13	Round 2 registration diagnosis (0 CU)	Raw STL → computational STL is a pure translation (error <0.1 mm); under `.xls` coordinates = raw STL coordinates, simulated mean is 0.113 (still 1/3 of wind tunnel), with 20/80 points within 1 grid cell of a wall; pre- vs. post-redevelopment cannot be determined from the repo alone
10-13 08:35	`ideas` ran AIJ Case E north wind	Completed with exit code 0 in 3 minutes, 2.21 CU; ~3× lower than wind tunnel (MAE 0.24/0.31, $r = 0.29/0.36$), registration unverified
10-13 00:16	Probed OpenCL on Colab A100 (without `high-mem`, 40 GB)	NVIDIA CUDA platform recognized A100, driver 580.82.07, `g++` 13.3 / `cmake` present, works out of the box; session closed, ~0.1 CU

Open Items and Next Steps

- **10-14 01:05**: Zero-compute re-check by `ideas`: after extending the upwind flat fetch, 55 of the 75 pure-fluid probes still decreased (MAE 0.128 → 0.144), showing the drop is not caused by mixed-cell interpolation points; however, `r5` and `r6b` had different averaging windows (141 vs. 256 s), so the difference cannot yet be attributed purely to fetch length. Decision: merge the window control into Round 7 (standardize `r5` domain to 31,000 steps / 20,000 averaging steps and output both S_{CFD} and S_{exp} metrics), holding the empty-flat-ground mechanism test until Round 7 results arrive (skip if mean ratio ≥0.9). Decision recorded.
- **10-13 21:50**: Job 1536823 exited with code 127 (batch shell lacked the `module` command, solver never started, no results produced; failed run root retained). Dispatched `ideas` to fix the startup environment in a new CRC root and re-run: **Job 1539640** queued (root `luw_aij_w270_dx1_r4crc_retry_20261013_2150`, ComputeOps request registered, ETA 10-14 07:52; duplicate platform dispatch 1539652 cancelled) (decision); submit for review together with Round 5 once completed.
- 10-13 11:50 Round 4 (Colab, west wind, 1 m, cropped domain 1200×700 m) was reclaimed by Colab at 29–36% across all three attempts, yielding no mean field and no reportable MAE (platform issue). Re-dispatched to CRC 2×A10: submitted Job 1536823 at 10-13 12:06 (resource request registered, expected completion before 22:06, wall-clock cap 6 h; CRC does not deduct SU, recorded as 0 SU on row 32 of the PETW SU ledger, allocation cap 2 GPUs × 6 h = 12 GPU·h).
- Assigned by Liang Hu on 10-13: use this project for a blind test of Gemini 4 Argon (`gpu-gemini`), providing a data package containing raw data only, with the Lead's prediction sealed in advance; score against Job 1536823 once results arrive and report back to Liang Hu with copies to relevant Leads. Argon has submitted its report; Lead's preliminary assessment notes two new leads—`setup.cpp` may shift buildings by 1 grid cell in output coordinates, and the wall immersed boundary described in the paper is absent from the source code—but Argon fabricated a variable name and misread the configuration of 1536823. Instructed `ideas` to empirically verify the 1-cell offset during 1536823 post-processing.
- 10-13 Supplementary material check: obtained the journal supplementary PDF (S1–S8); S1 covers Guangzhou radar Kriging boundaries and does not contain the domain or U_{ref} for AIJ Case E; S4–S5 (PDF pp. 6–10) state that buildings and ground use an immersed boundary with a "three-point regularized delta kernel + multi-direct forcing." **Lead verification: none of the three solver directories in the public code (`FluidX3D`, `FluidX3D-3.5`, `CULBM`) contain this implementation**—walls use voxel bounce-back—and `LUW-v2.8` is absent from the public repository. Thus 1536823 represents at best a reproduction of "public code + paper settings," not a reproduction of the code used in the paper.
- 10-13 19:40 Official AIJ materials verified (see Parameter Source Table): the official comparison domain is only ~500 m, whereas the LUW example adds ~800 m of flat ground on all four sides; the STL represents the "post-redevelopment" geometry; the wind tunnel measures scalar wind speed. Dispatched Round 5: run west wind at 1 m on the official domain (Colab, ≤8 CU).
- 10-13 Retrieved the Gemini report; Lead spot-checked two findings and confirmed both (integer multiples in `.xls`; Yoshie 2007 definition). **Comparisons are standardized to the "post-redevelopment, west-wind column,"** matching Han 2026.

- **10-13 19:41 Round 5 (official AIJ scale 500×500 m domain, west wind, 1 m, Colab 1.11 CU)**: against post-redevelopment west wind, 80-point MAE is 0.148 (unshifted) / **0.1425 (with the $(+1,+1,0)\text{ m}$ offset confirmed by the solid mask)**, $r = 0.80$, simulation/wind-tunnel mean ratio 0.81; pre-redevelopment MAE is 0.20. Numerically comparable to 0.143 in the paper. The solid mask confirmed that output coordinates are shifted by $+1$ cell in both x and y (unshifted in z) with IoU 1.000—confirming Argon's `setup.cpp` offset finding in x and y. Upwind 2 m wind speed drops from 105% at the inlet to 43% at 7 m ahead of the buildings, but that location sits right against a building face and is affected by blockage, so it cannot be cited as evidence of flat-fetch decay. **Not yet submitted for review and not termed a successful reproduction**; hold until 1536823 (same wind direction, 509 m upwind flat fetch) finishes and submit both together.
- 10-13 Job 1536823 failed (exit code 127: job script did not initialize `module`, solver never launched), old run root retained; resubmitted from a new run root at 10-13 21:52 as **Job 1539640** (supplemental ComputeOps request registered, expected completion before 10-14 07:52), adding `module` initialization and startup self-checks. Job 1539640 then failed during pre-check (exit code 82: GPU `minor_number` query failed) before the solver launched; both failed roots retained. Re-dispatched as Round 6: take the Round 5 500 m domain and extend only the upwind edge to a 509 m flat fetch (single-variable control) on a single GPU—trying Colab first and falling back to a single CRC GPU if reclaimed.
- 10-14 00:xx Round 6 completed; the near-ground decay lead is **retained** (over the flat fetch, 2 m wind speed retains only 33–53%, dropping within 50–150 m of the inlet because the ground boundary cannot sustain the profile); extending the upwind flat fetch changes MAE by only 0.015. Sent to 6 reviewers (with evidence package); Argon's evaluation has been written and copied to the Platform Lead.
- Round 4 running (`ideas`). Once results return, decide whether to retain or retract the near-ground decay lead (Root Lead has already notified the sister line and sent an interim note "do not cite as corroborating evidence yet"), whether to shelve or continue, and inform the Root Lead whether a further update to the sister line is needed.
- Intra-line rule (Root Lead 10-13, trial through 10-20): routine fixed-questionnaire Q&A chores default to the lightweight channel with 10% spot-checking; this project currently has no such chores.
- When delegating web-search / download tasks to `gemini-worker` (until the timeout wrapper is fixed): include "append the file inventory to `REPORT_PARTIAL.md` after every downloaded file" in the brief and break tasks into ≤30-minute chunks.
- Funding classification (License Clause 8) remains a pending item for Liang Hu on the Digital Twin Wind Tunnel line and will not be queried separately here.

Literature Basis

- Han 2026 literature card (see above), pp. 38–43

Notes from Liang Hu

- 10-14 Liang Hu: Judging from the comparison curves, neither the wind-speed metric definition nor the averaging window is the crux; the ground boundary layer is important for the lower atmosphere as well—asked whether it can be added subsequently.
- 10-14 Liang Hu: Complete all configuration-level modifications (inflow turbulence, mass-flux correction, averaging window, domain height) first before deciding on source-code changes or contacting the authors; he considers code version differences unlikely to cause a gap of this magnitude—prioritize setup differences first, leaving version and immersed-boundary questions for later.

Cross-Line Note (10-13, Correction from Digital Twin Wind Tunnel Lead)

- What decays in LUW Case E is near-ground **mean wind speed**; in Silsoe v7, near-ground mean wind is maintained along the fetch (~4.2–4.3 m/s at $z = 1\text{ m}$ vs. ~4.33 from the log law), whereas what decays is **turbulence structure** (I_w increases while L_u/L_w decrease). Therefore, for v7 our finding serves only as weak corroborating evidence that "FluidX3D-family codes struggle to maintain near-ground turbulence," whereas its symptoms are much closer to Urban Field T3/T4 (where mean wind drops to 0.81–0.82 of the log law). Cite under this exact framing if this project is restarted.

Key figures